FY26 Spillway Drainage Inspection
Garrison Dam, ND

Attachment 1 – Reference Drawings

For Information Only

1. Spillway Subdrain, Relief Wells, and Box Drain Figures

Figure 1 through Figure 5 provide typical figures and photographs of the spillway subdrain, crest relief wells, and box drains.

Figure 6 provides an overview map of the spillway drainage features.

Figure 7 presents a summary of the spillway subdrain inspections and repairs completed after the 2011 flood through 2016. The figure includes tables summarizing the sediment collection per segment, drain defects observed per segment, and notes regarding the relief well and crest leader pipes along Row 1. Cured-in-place pipe repairs were subsequently completed in 2012 and 2014 at the highest priority pipe fractures, and these locations are presented on the map.

Figure 8 presents the drainage measurements in the spillway chute subdrain and box drains. The measurements at MH-W1 include inflows from the crest subdrain leaders and relief wells.

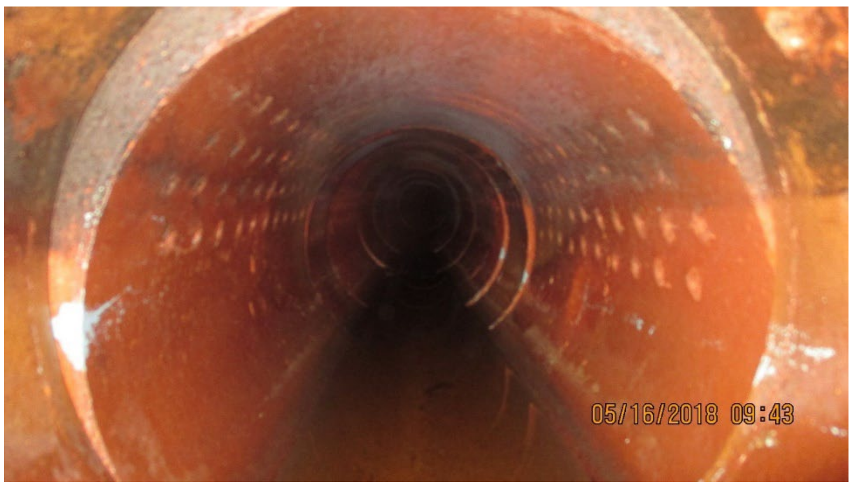
Figure 9 through Figure 12 provide additional details regarding the Row 1 drainage features. Photographs are provided of the sandbag ring and pumping needed to access Row 1 manholes for a camera inspection in 2025. The inspection was conducted by USACE without any flushing or cleaning of the pipes prior to operating the camera. Additional photos show the locations where there is currently gravel and sediment identified along the Row 1 alignment.

Figure 13 through Figure 22 provide original construction drawings for the spillway drainage features.

2. Powerhouse Drainage Figures

Figure 23 through Figure 26 provide additional maps of the powerhouse drainage system and flow readings.

Example manhole, MH-W9 facing upstream segment.



Example lateral pipe, MH-W9 facing east inflowing segment.

Example lateral pipe, MH-C9 facing east segment.

Figure 1. Example subdrain inspection photos.

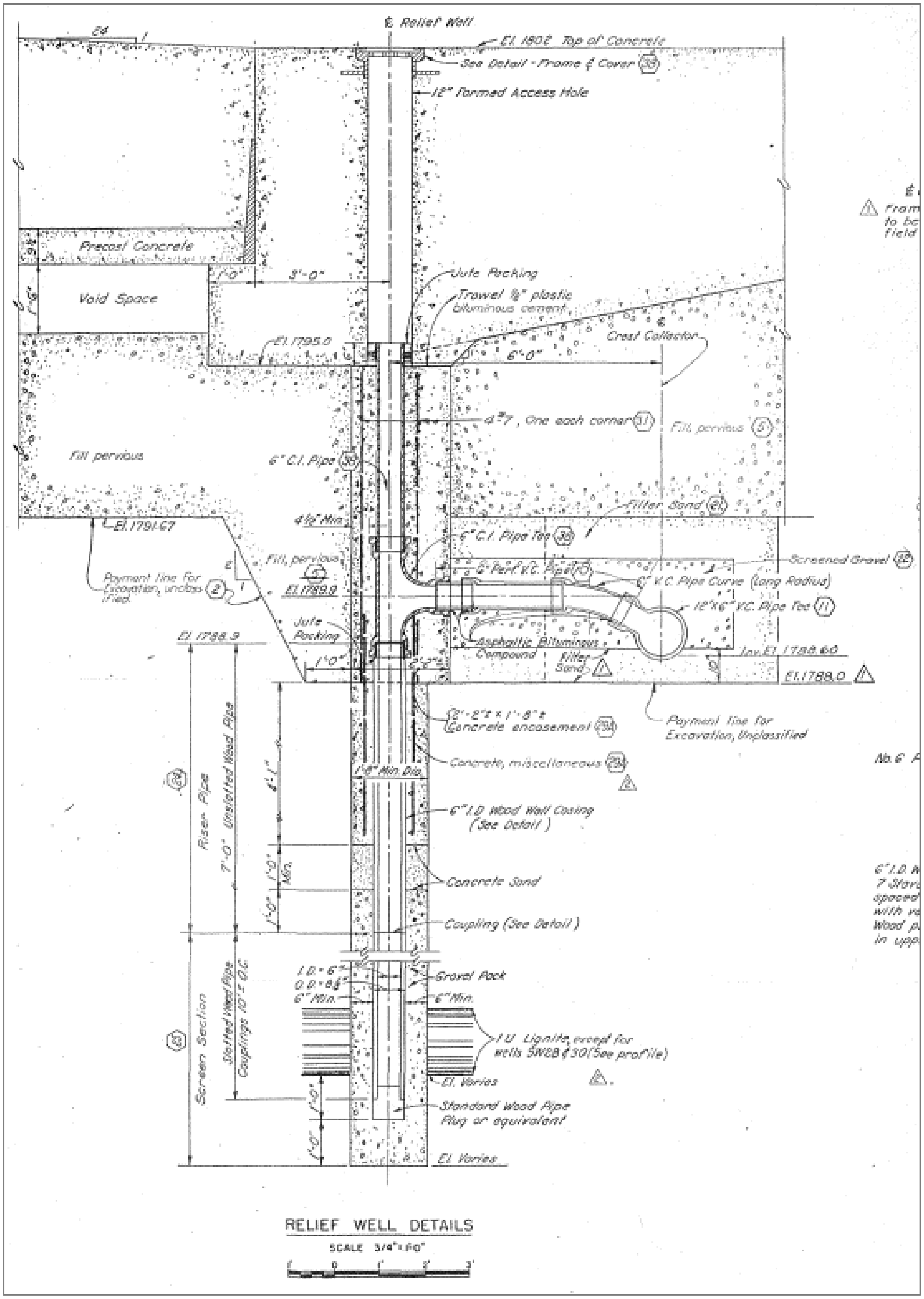


Figure 2: Crest Relief Well Detail

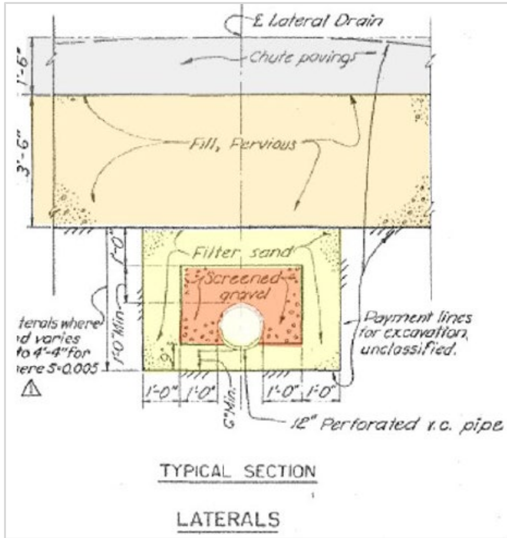


Figure 3. Chute subdrain lateral drain typical detail.

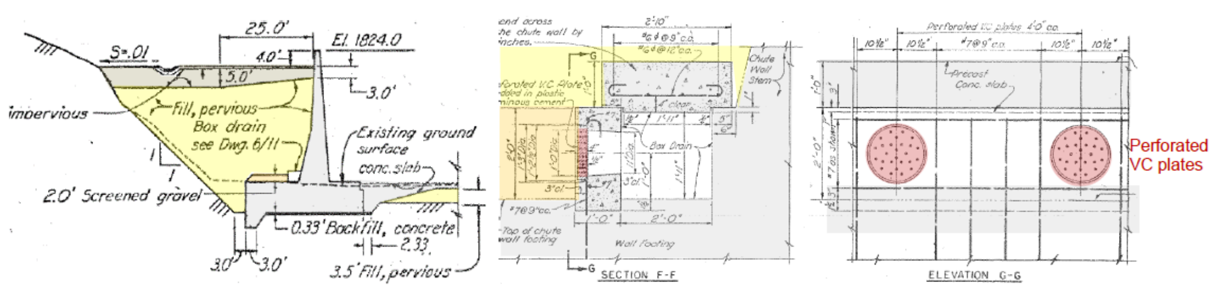


Figure 3-30: Typical box drain section and profile

Drawing ref: GS4-6/6 and GS1-6/17

Figure 4: Box drain section and typical profile

c) Box drain camera inspection, typical view.
d) Box drain typical perforated flow inlet.

Figure 5: Box drain typical inspection images

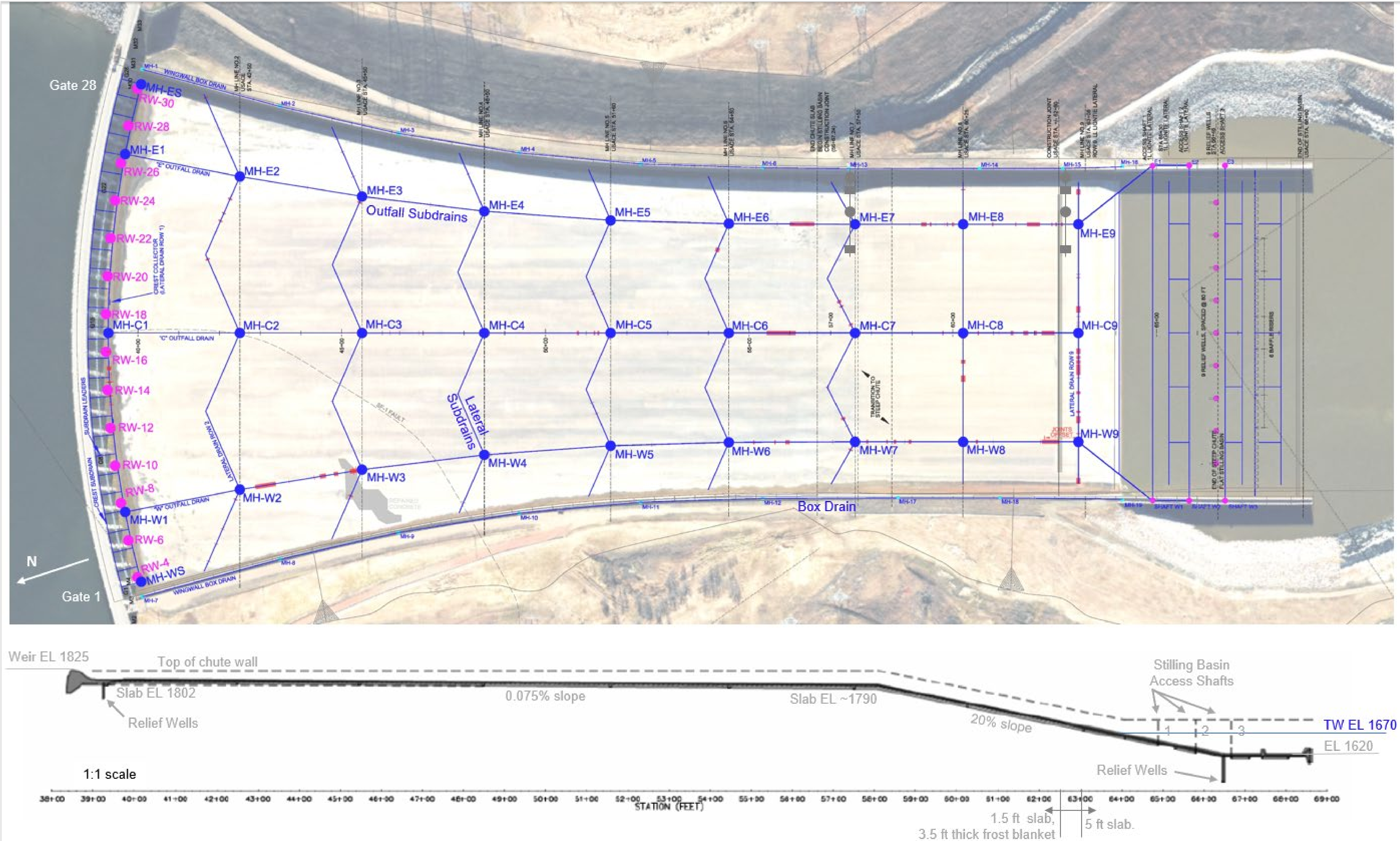
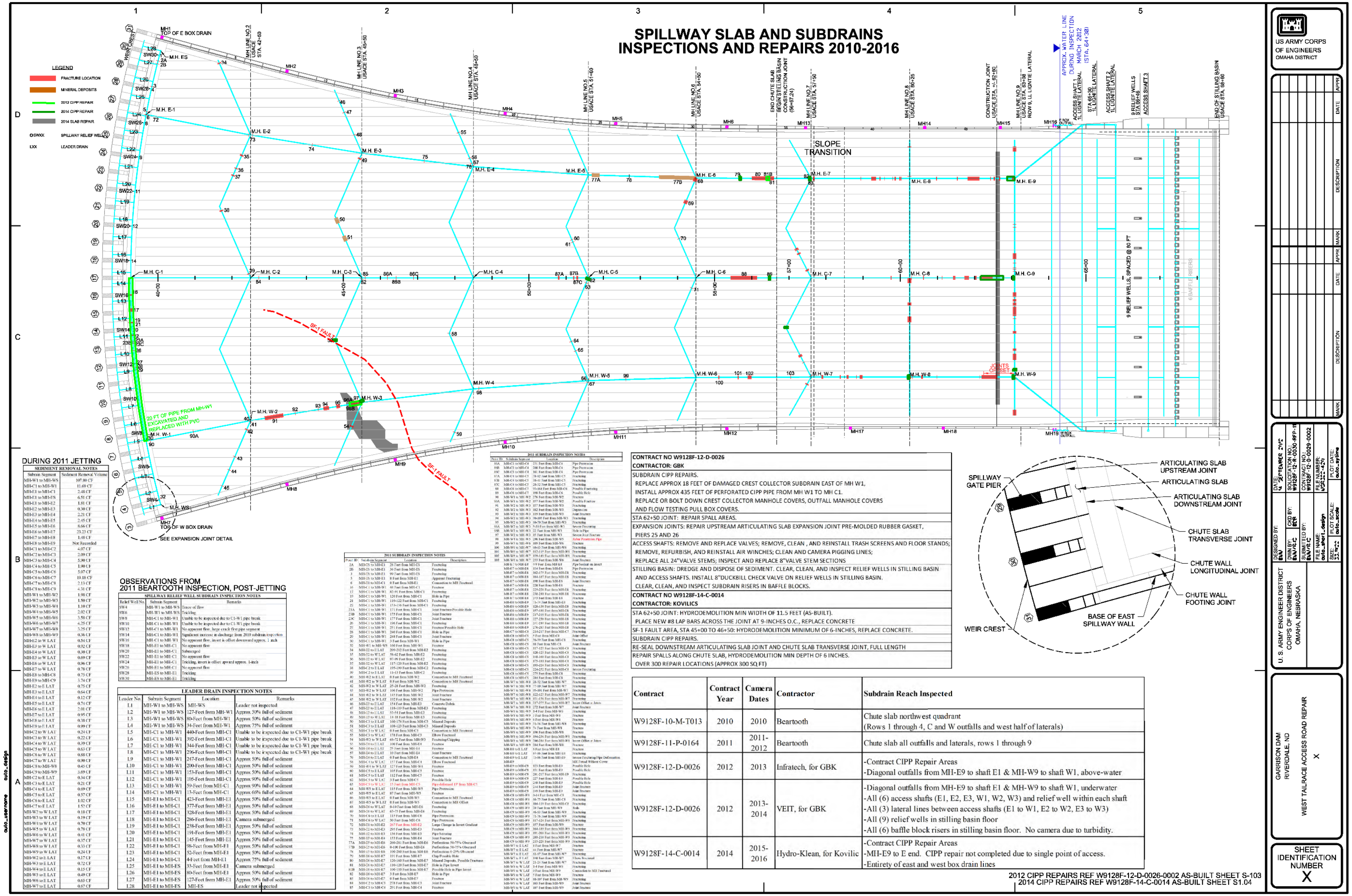


Figure 6. Drainage overview map.



Contract	Contract Year	Camera Dates	Contractor	Subdrain Reach Inspected
W9128F-10-M-T013	2010	2010	Beartooth	Chute slab northwest quadrant (Rows 1 through 4, C and W outfalls and west half of laterals)
W9128F-11-P-0164	2011	2011-2012	Beartooth	Chute slab all outfalls and laterals, rows 1 through 9
W9128F-12-D-0026	2012	2013	Infratech, for GBK	-Contract CIPP Repair Areas -Diagonal outfalls from MH-E9 to shaft E1 & MH-W9 to shaft W1, above-water
W9128F-12-D-0026	2012	2013-2014	VEIT, for GBK	-Diagonal outfalls from MH-E9 to shaft E1 & MH-W9 to shaft W1, underwater -All (6) access shafts (E1, E2, E3, W1, W2, W3) and relief well within each shaft -All (3) lateral lines between access shafts (E1 to W1, E2 to W2, E3 to W3) -All (9) relief wells in stilling basin floor -All (6) baffle block risers in stilling basin floor. No camera due to turbidity.
W9128F-14-C-0014	2014	2015-2016	Hydro-Klean, for Kovilic	-Contract CIPP Repair Areas -MH-E9 to E end. CIPP repair not completed due to single point of access. -Entirety of east and west box drain lines

Figure 7. Summary of Spillway drainage inspections and repairs, 2011-2016.

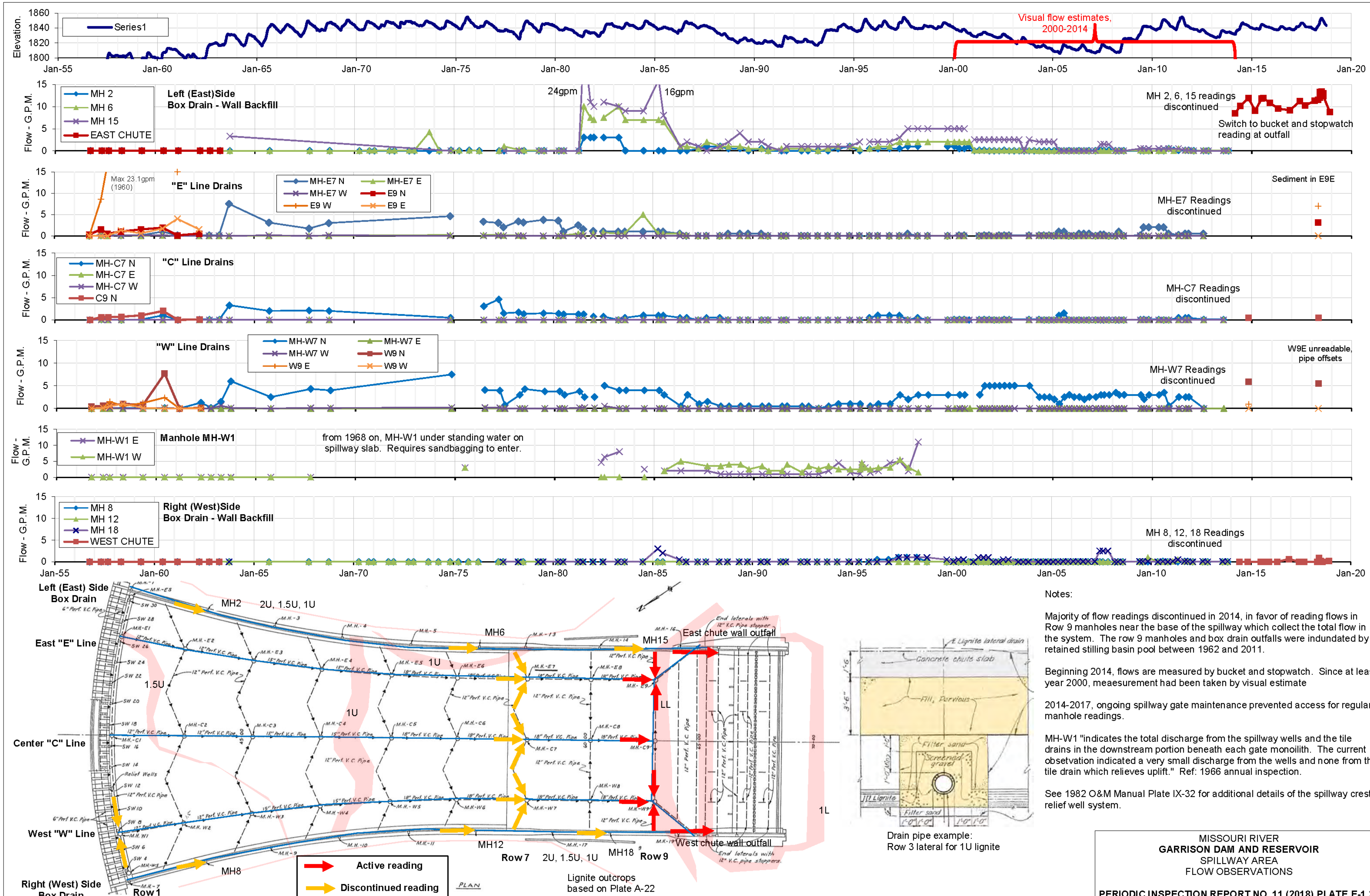


Figure 8. Drainage Measurements - Spillway

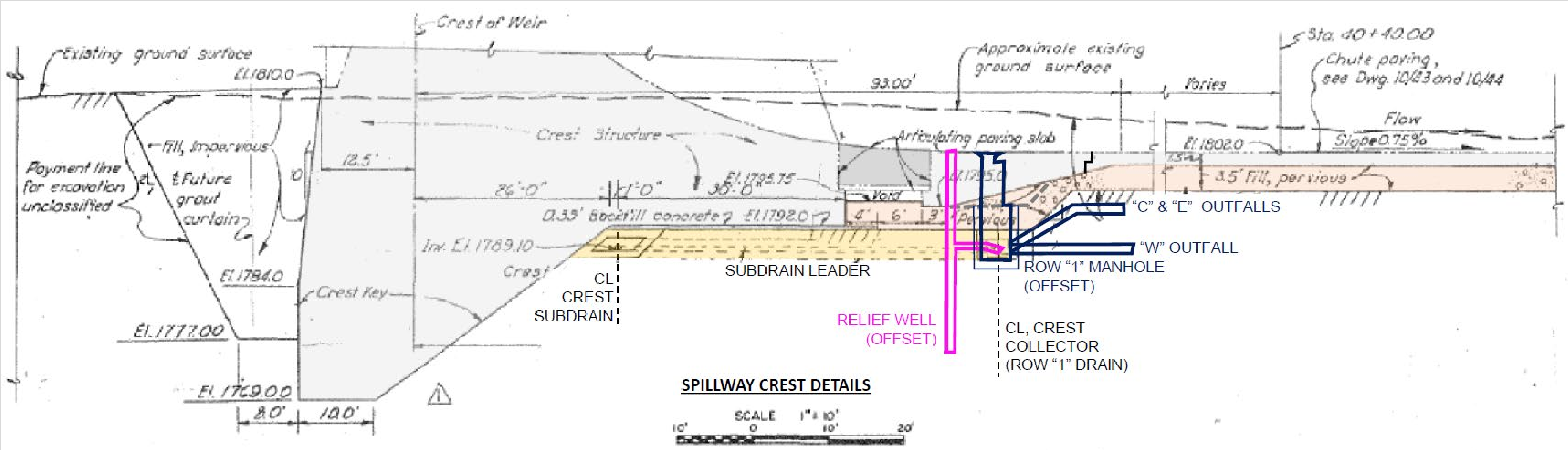
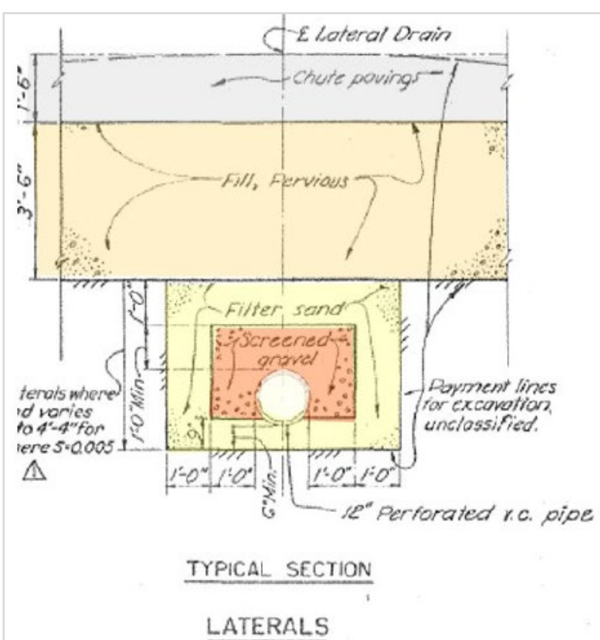
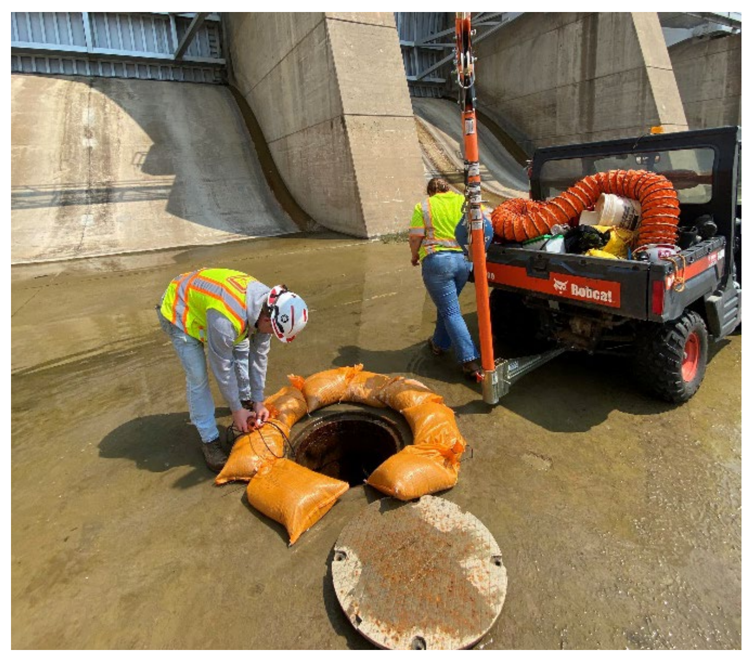


Figure 9. Spillway crest profile drainage detail.



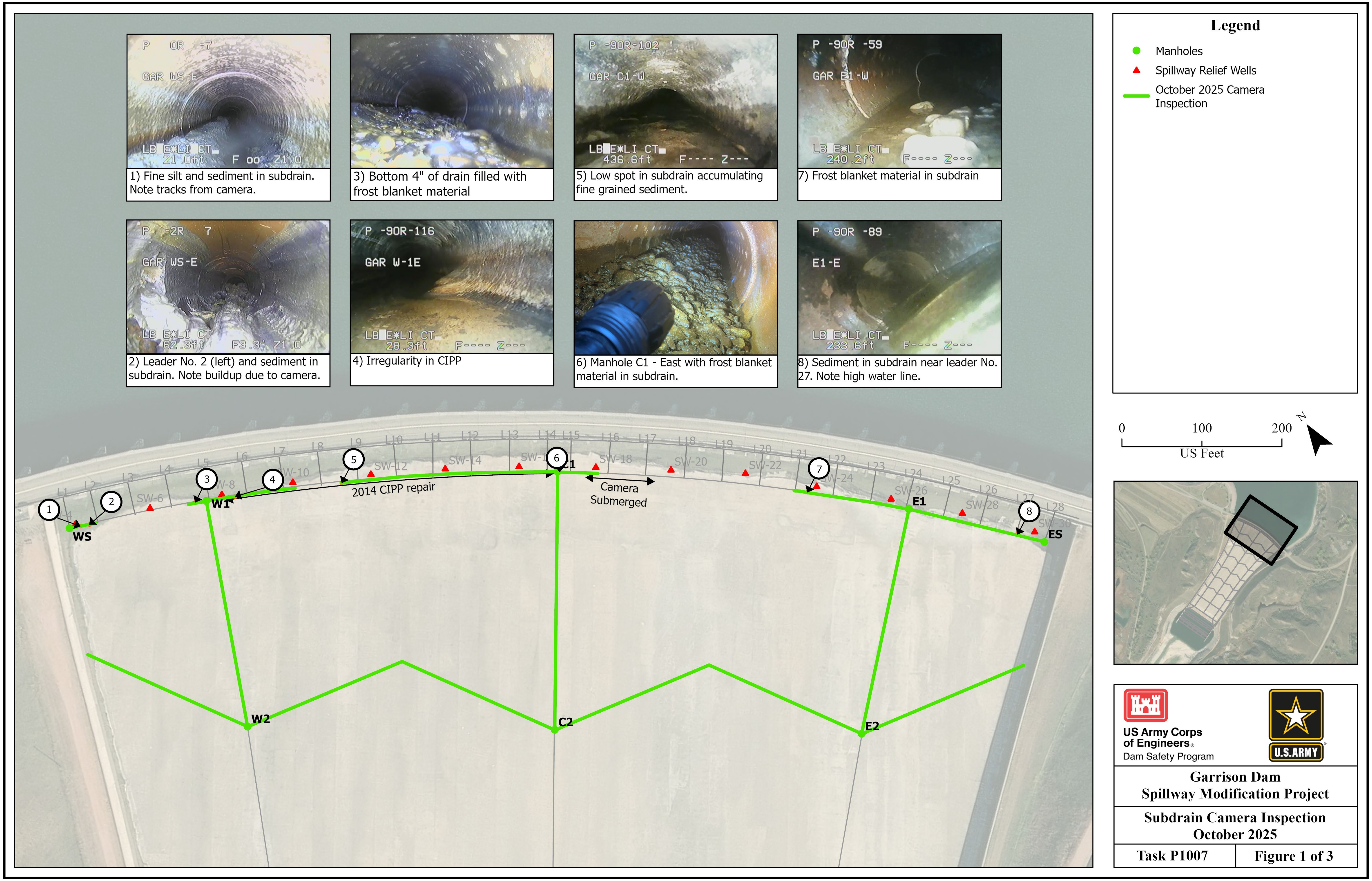
Lateral drain pipe typical detail



Row 1 manhole sandbag ring for access

Row 1 manhole sandbag ring for access

Figure 10. Lateral drain diagram and Row 1 access photos



2025 Camera Inspection Overview

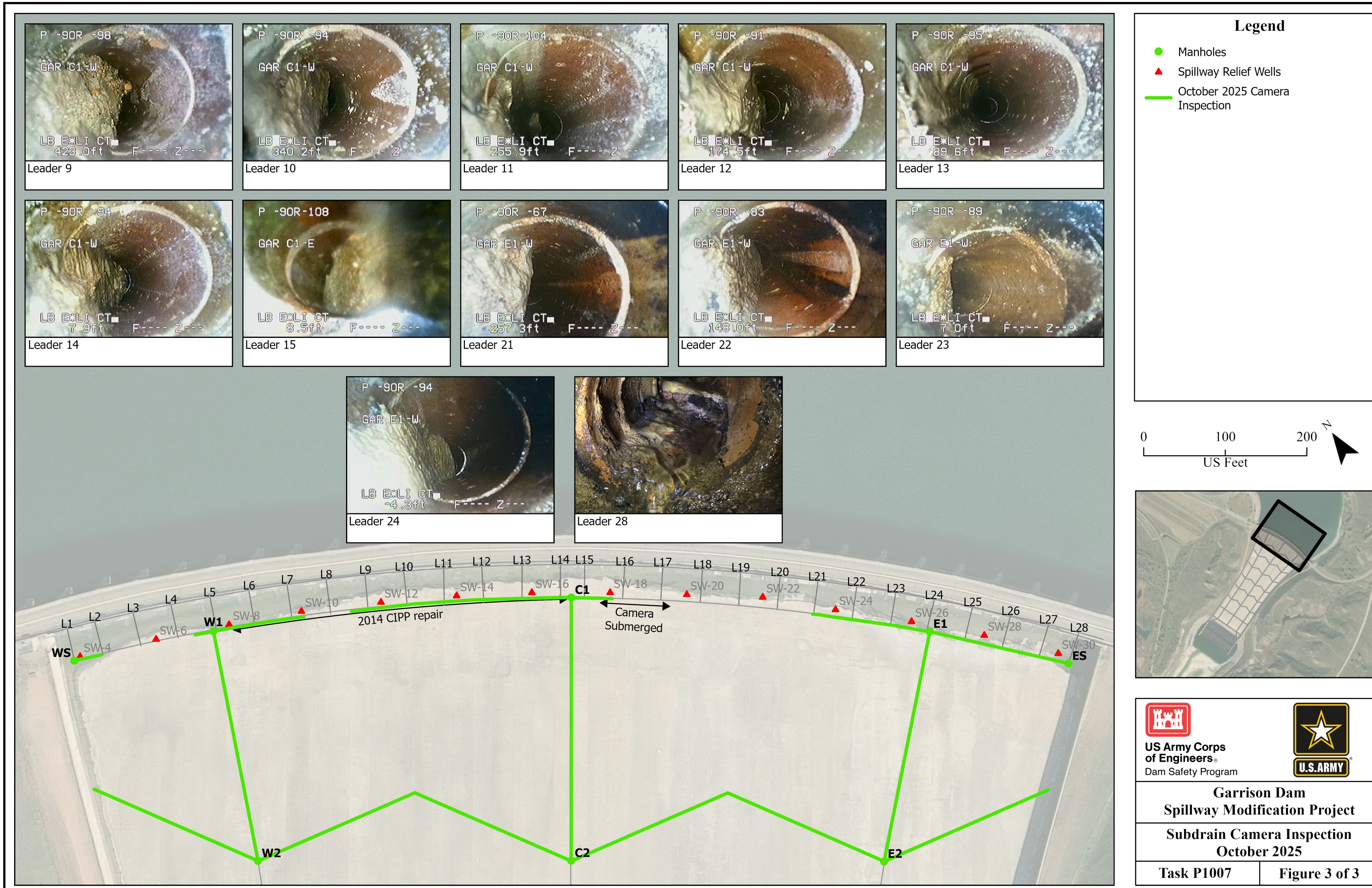
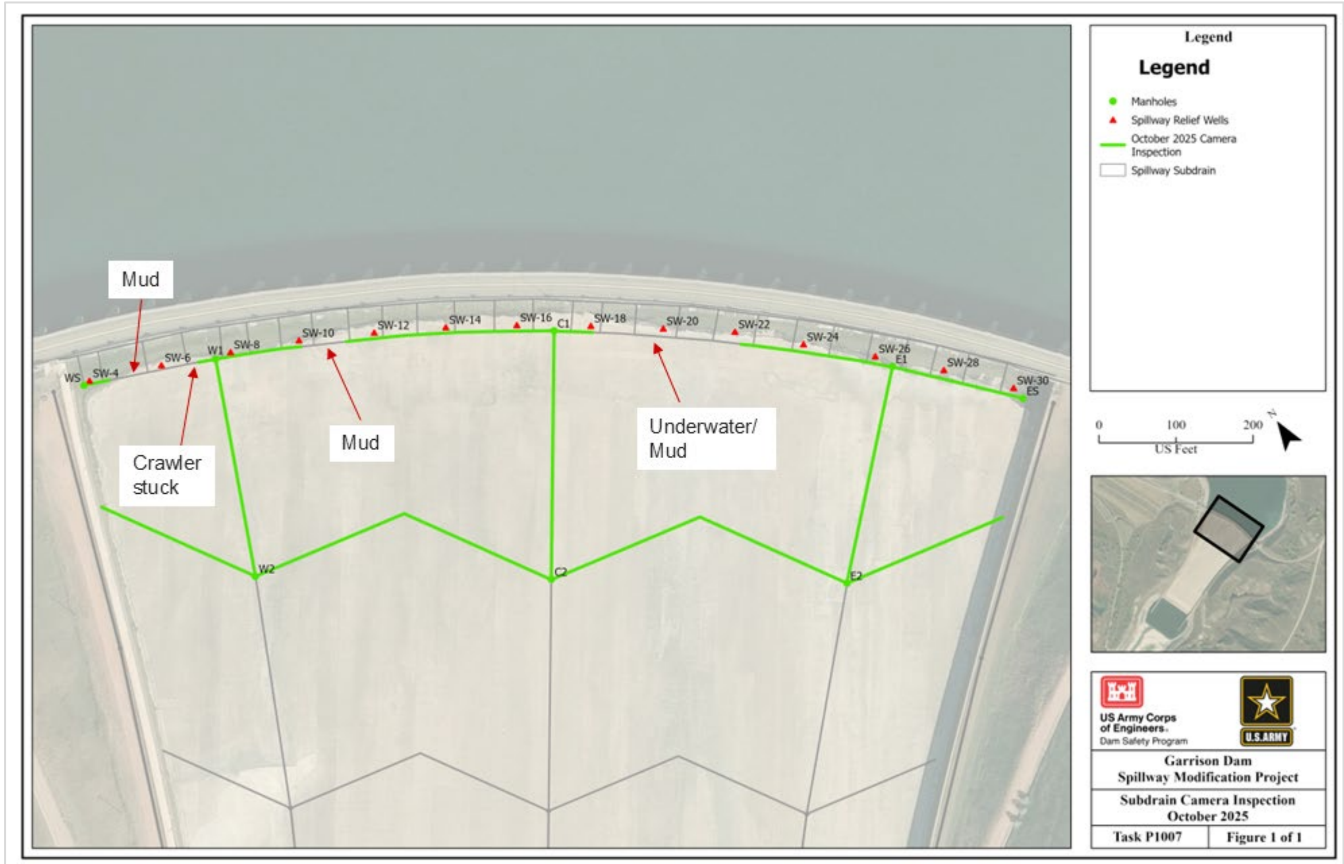


Figure 11. Crest Leaders Condition Summary, 2025.



Summarized 2025 camera inspection summary (no flushing performed)

Drainage gravel present in a Row 1 manhole.

Pumping dry a Row 1 manhole for inspection

Fine sediment and silt in a dry pipe segment

Drainage sand and gravel in a partially inundated subdrain location.

Figure 12. Additional 2025 Row 1 Inspection Map and Photos

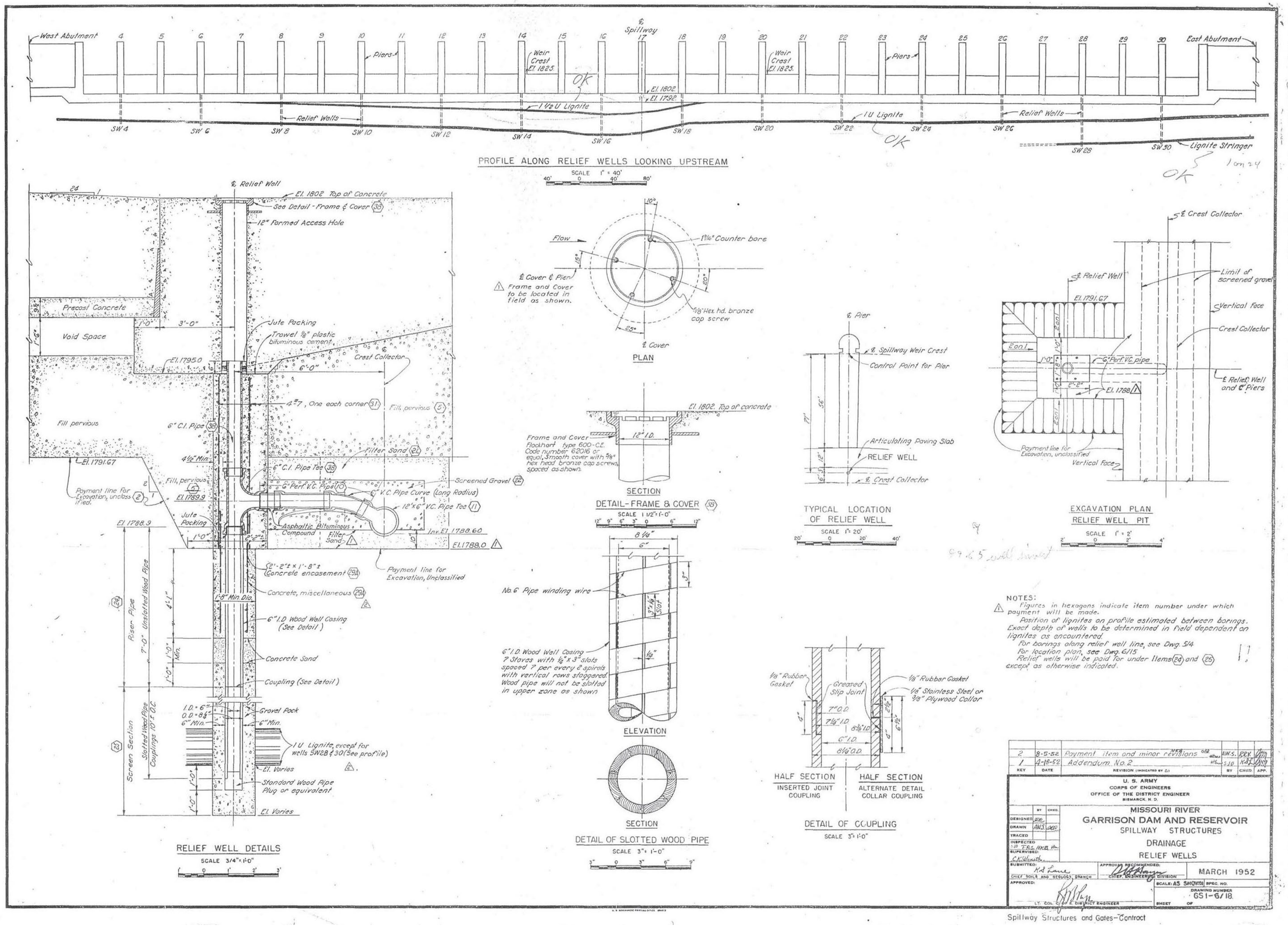


Figure 13. Crest relief wells original drawing (GS1, upper chute).

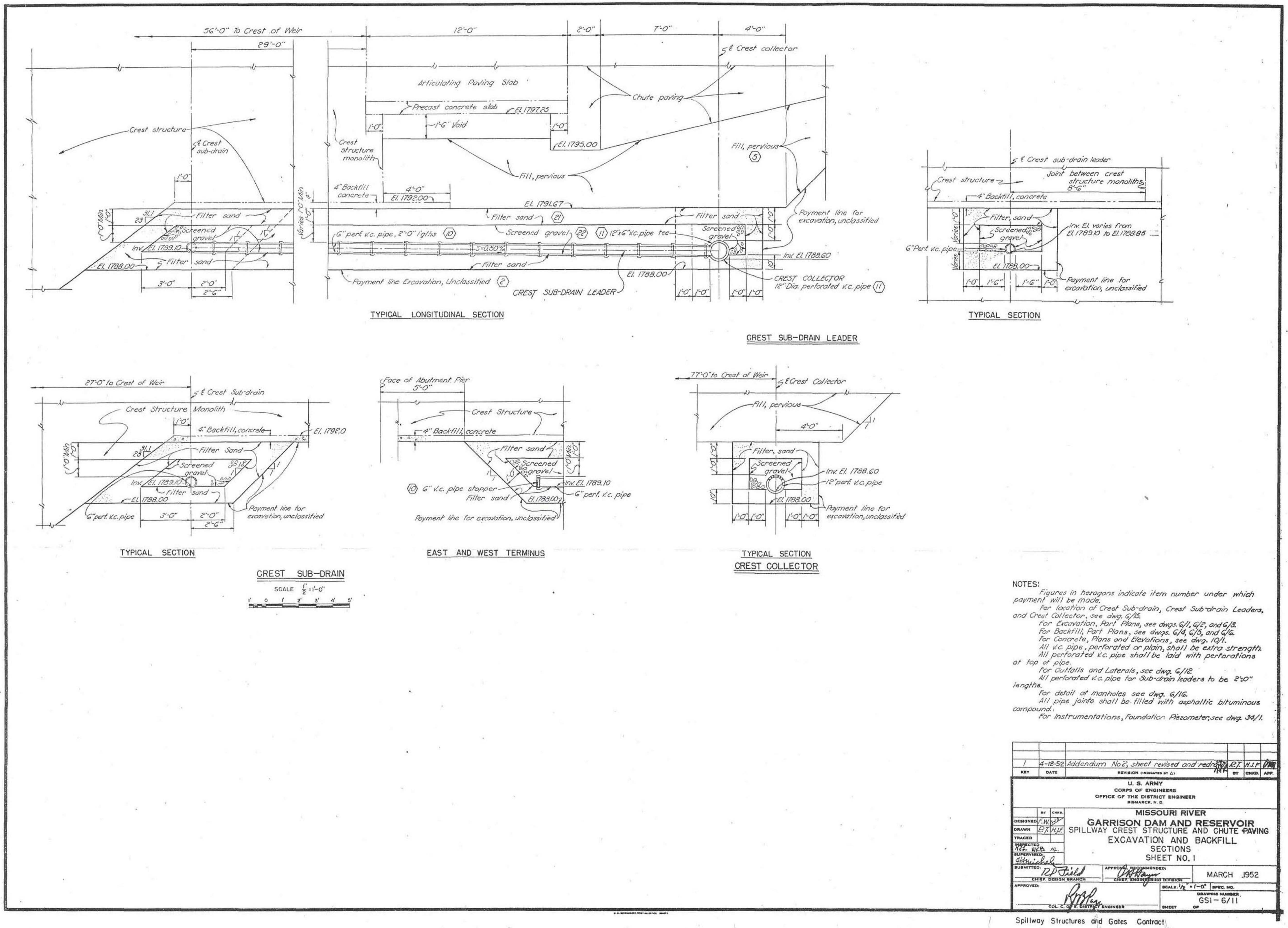


Figure 14. Spillway crest drainage original drawing (GS1, upper chute).

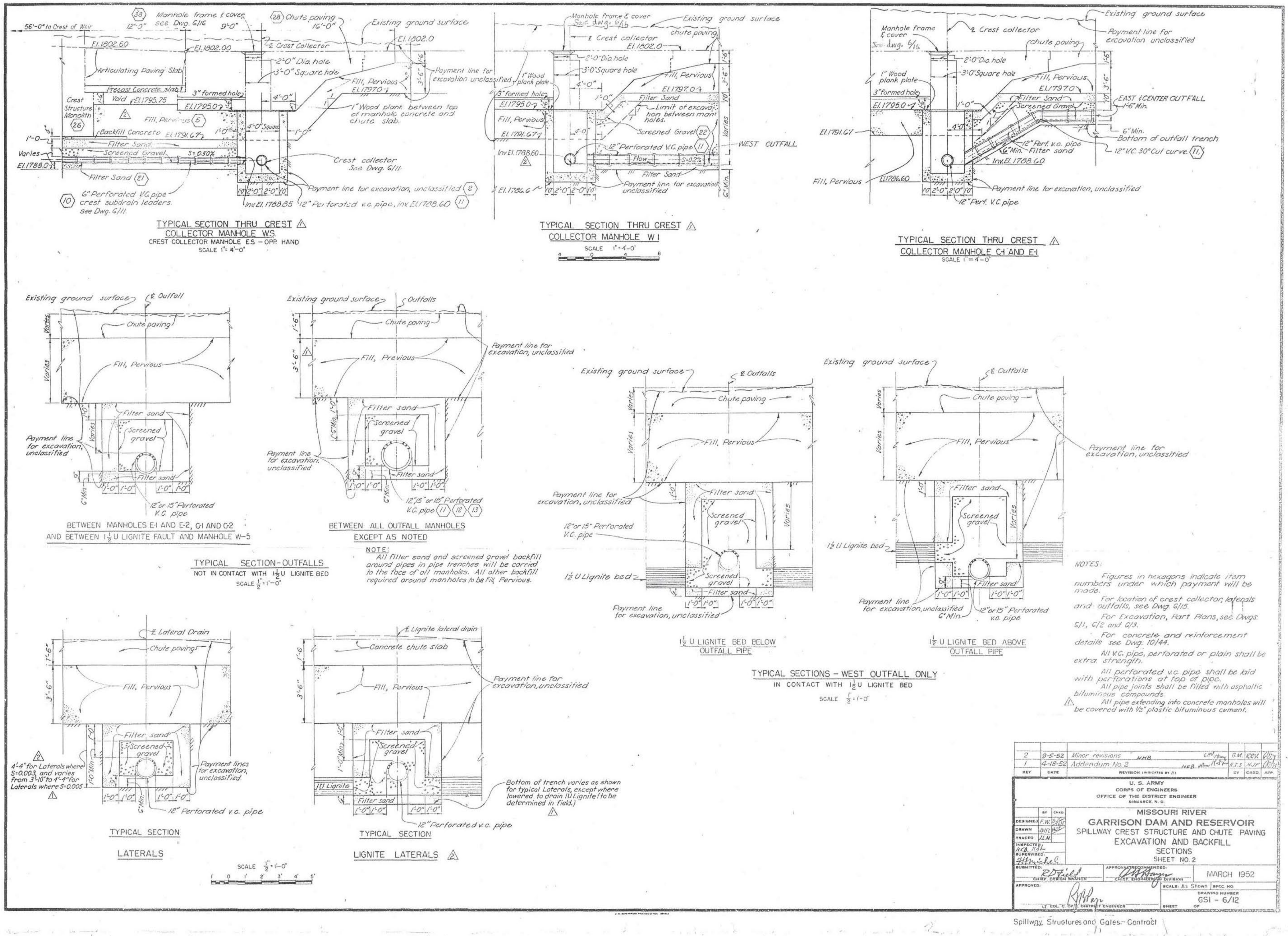


Figure 15. Row 1 manholes and typical lateral drains original drawing (GS1, upper chute)

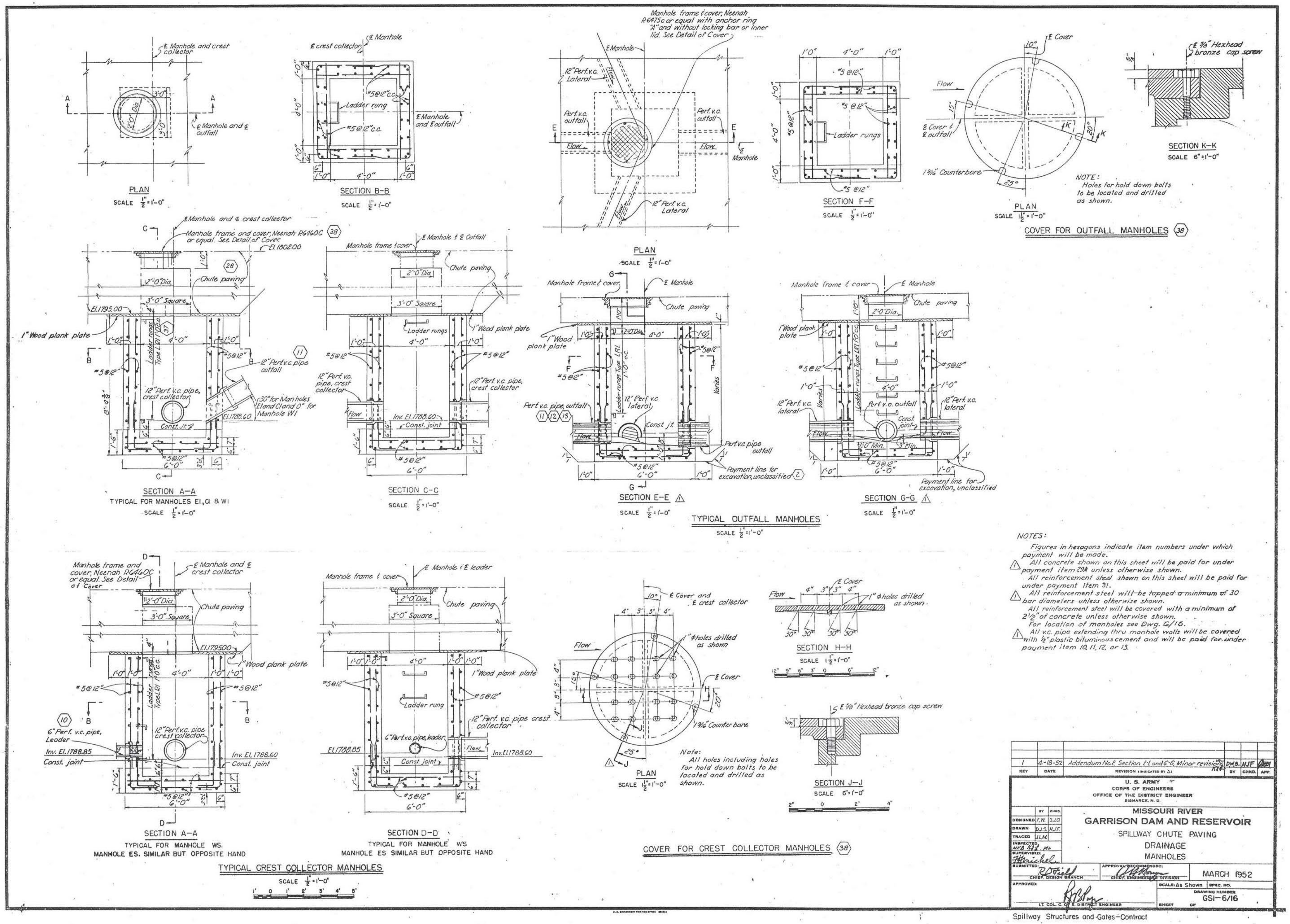


Figure 16. Spillway slab manhole details original drawing (GS1, upper chute)

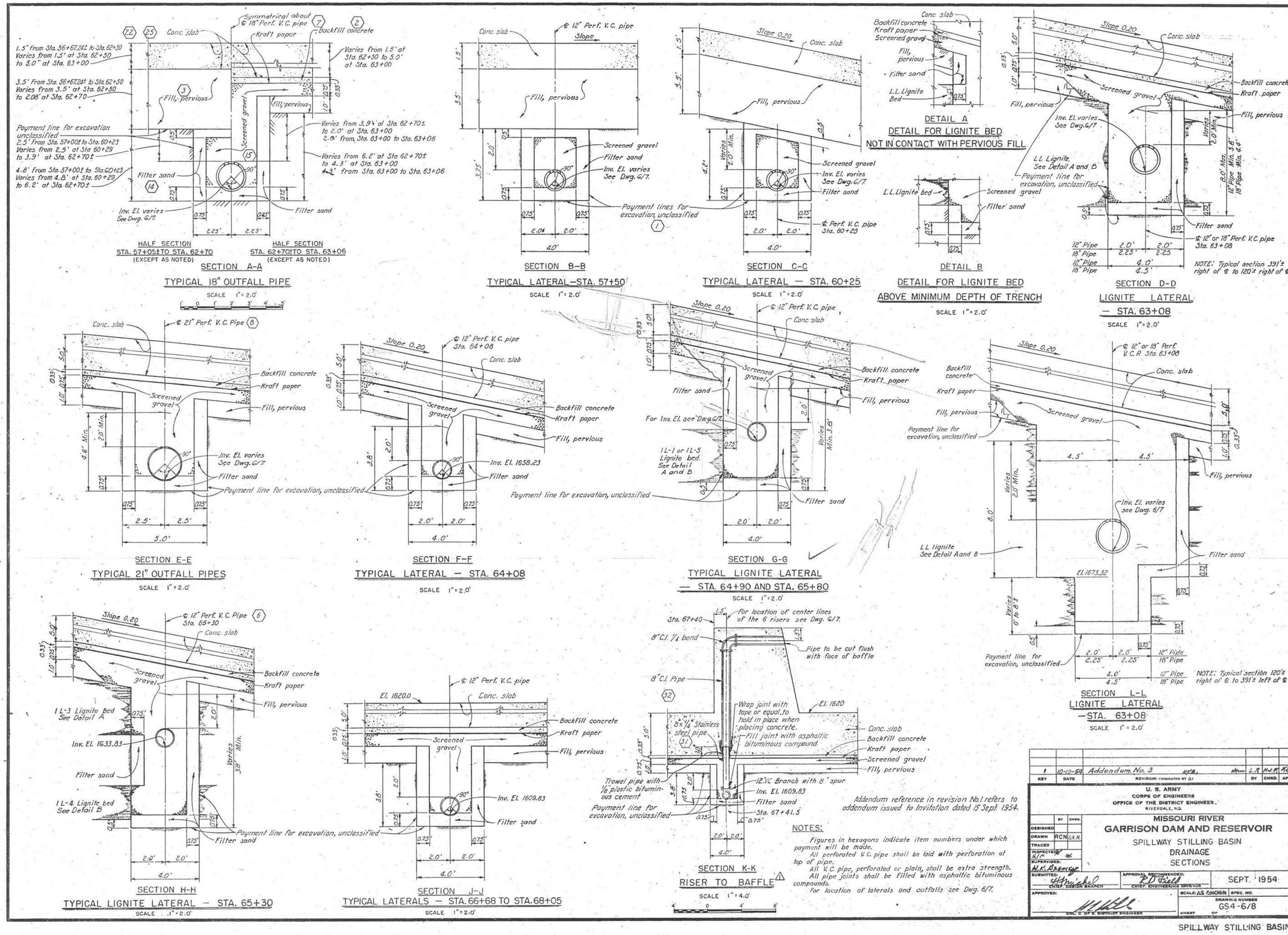


Figure 17. Spillway slab lateral drains original drawing (GS4, lower chute).

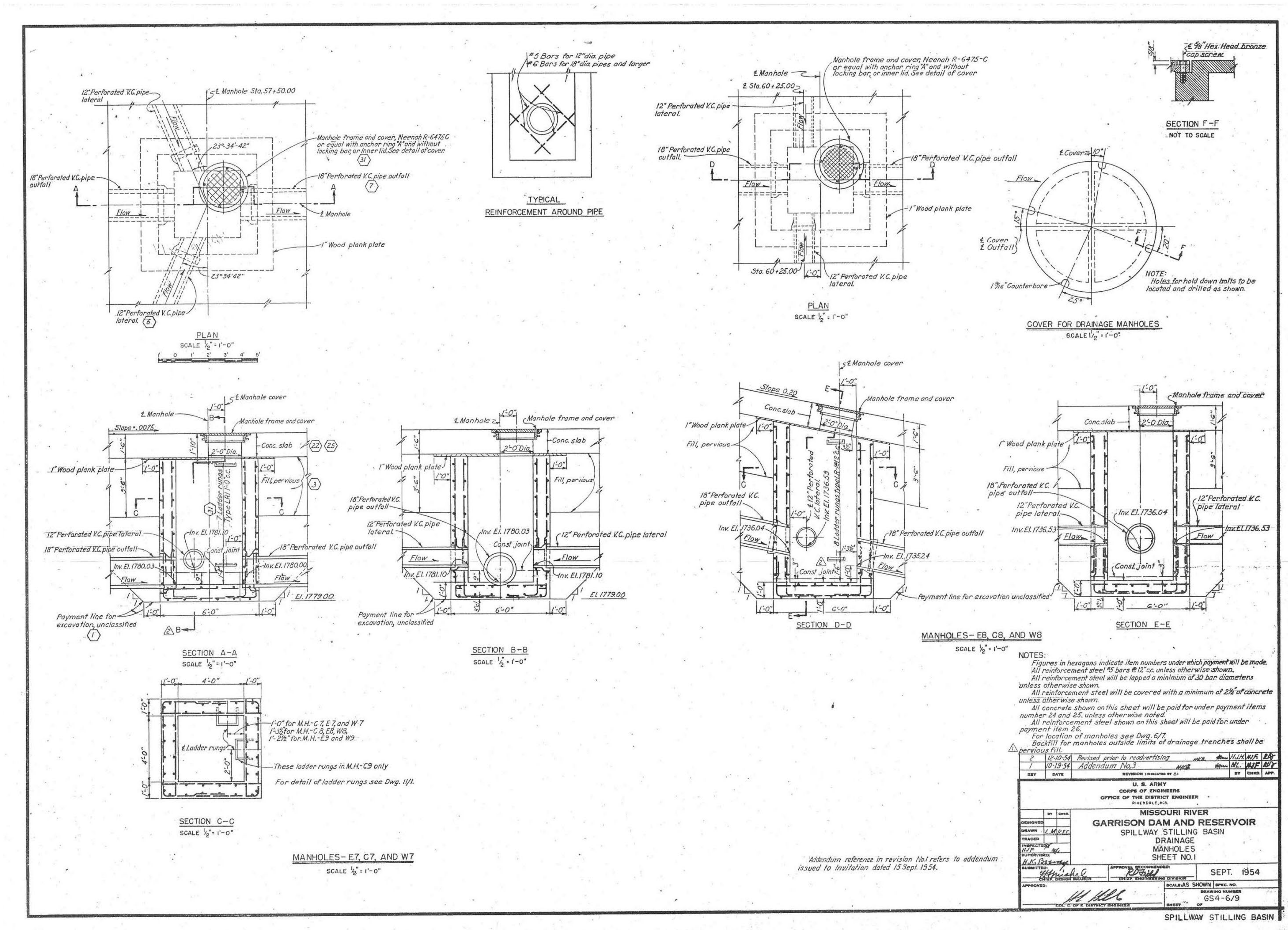


Figure 18. Spillway slab manholes original drawing (GS4, lower chute).

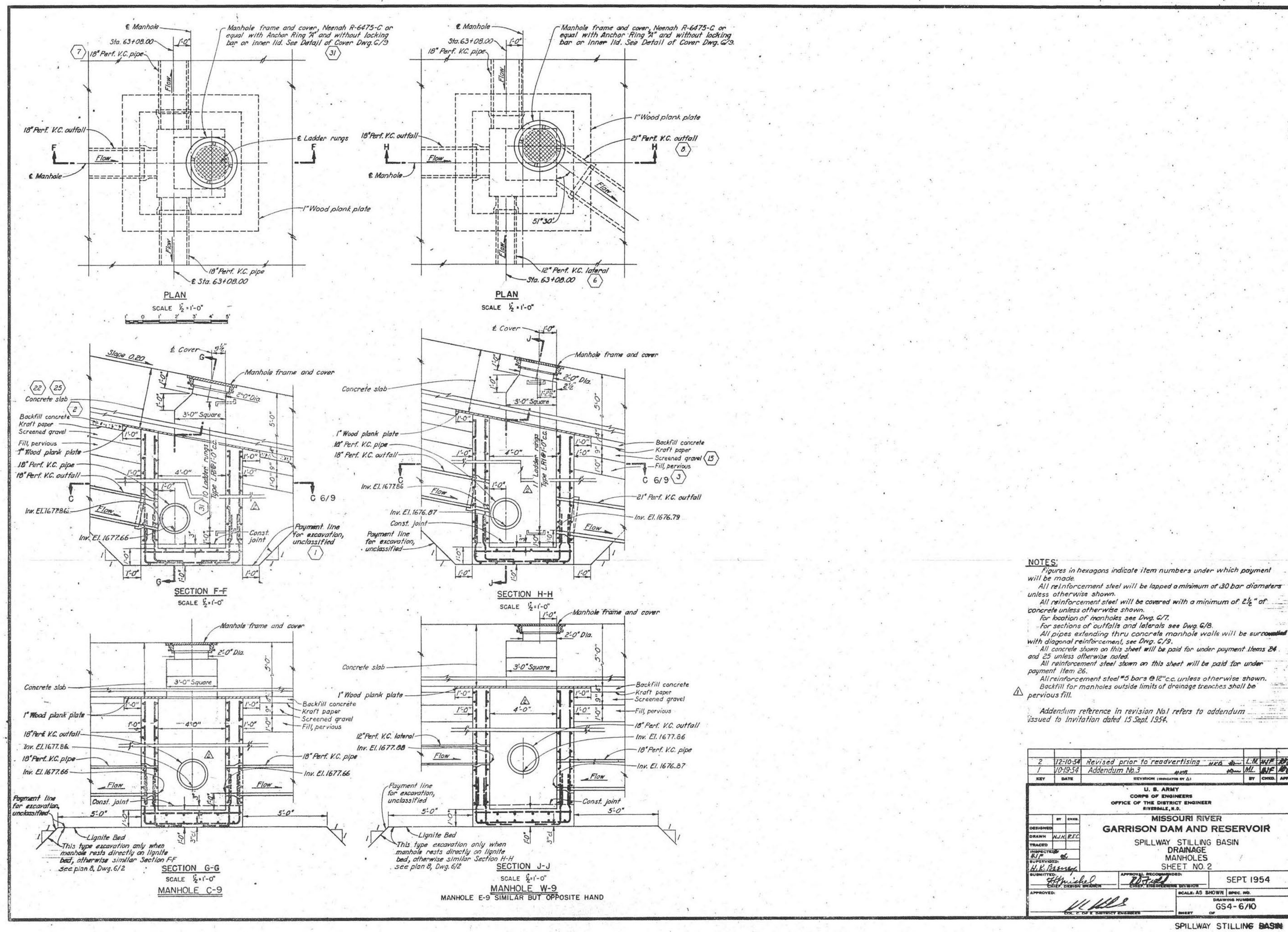
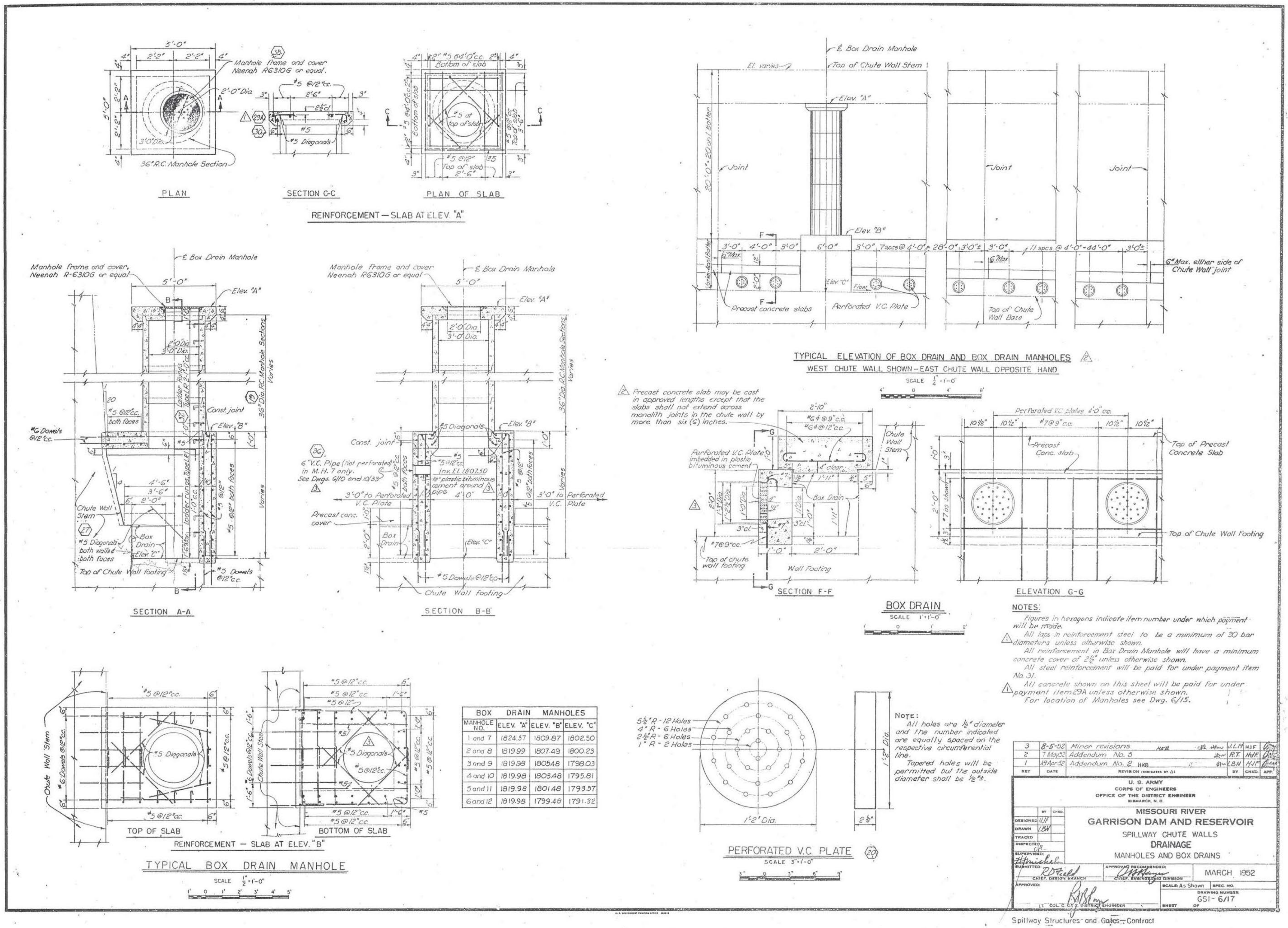


Figure 19. Spillway slab manholes original drawing (GS4, lower chute).



BOX DRAIN MANHOLES			
MANHOLE NO.	ELEV. "A"	ELEV. "B"	ELEV. "C"
1 and 7	1824.37	1809.87	1802.50
2 and 8	1819.99	1807.49	1800.23
3 and 9	1819.98	1805.48	1798.03
4 and 10	1819.98	1803.48	1795.81
5 and 11	1819.98	1801.48	1793.57
6 and 12	1819.98	1799.48	1791.32

Figure 20. Box drain profile detail (GS1, upper chute)

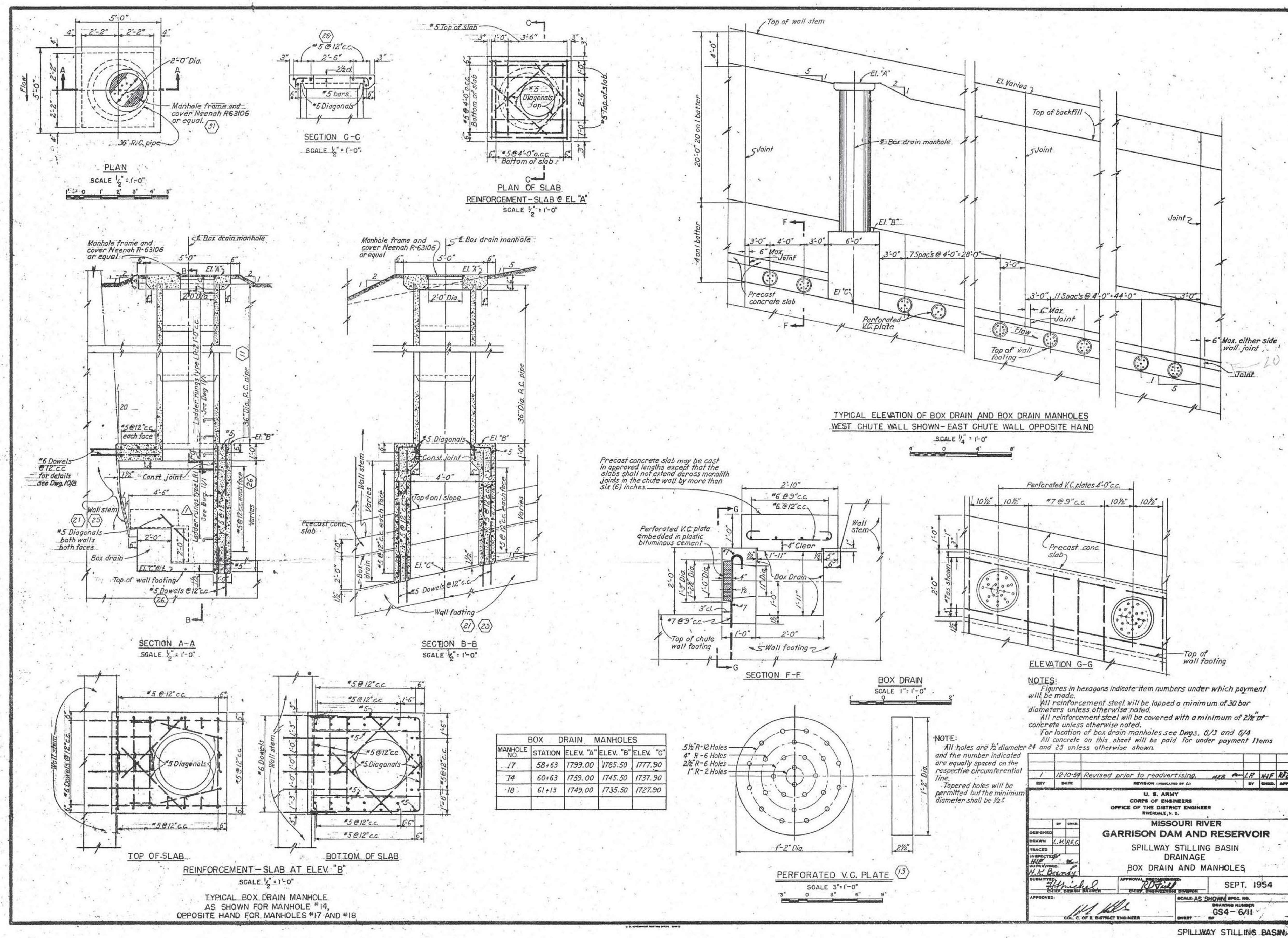


Figure 21. Box drain profile original drawing (GS4, lower chute).

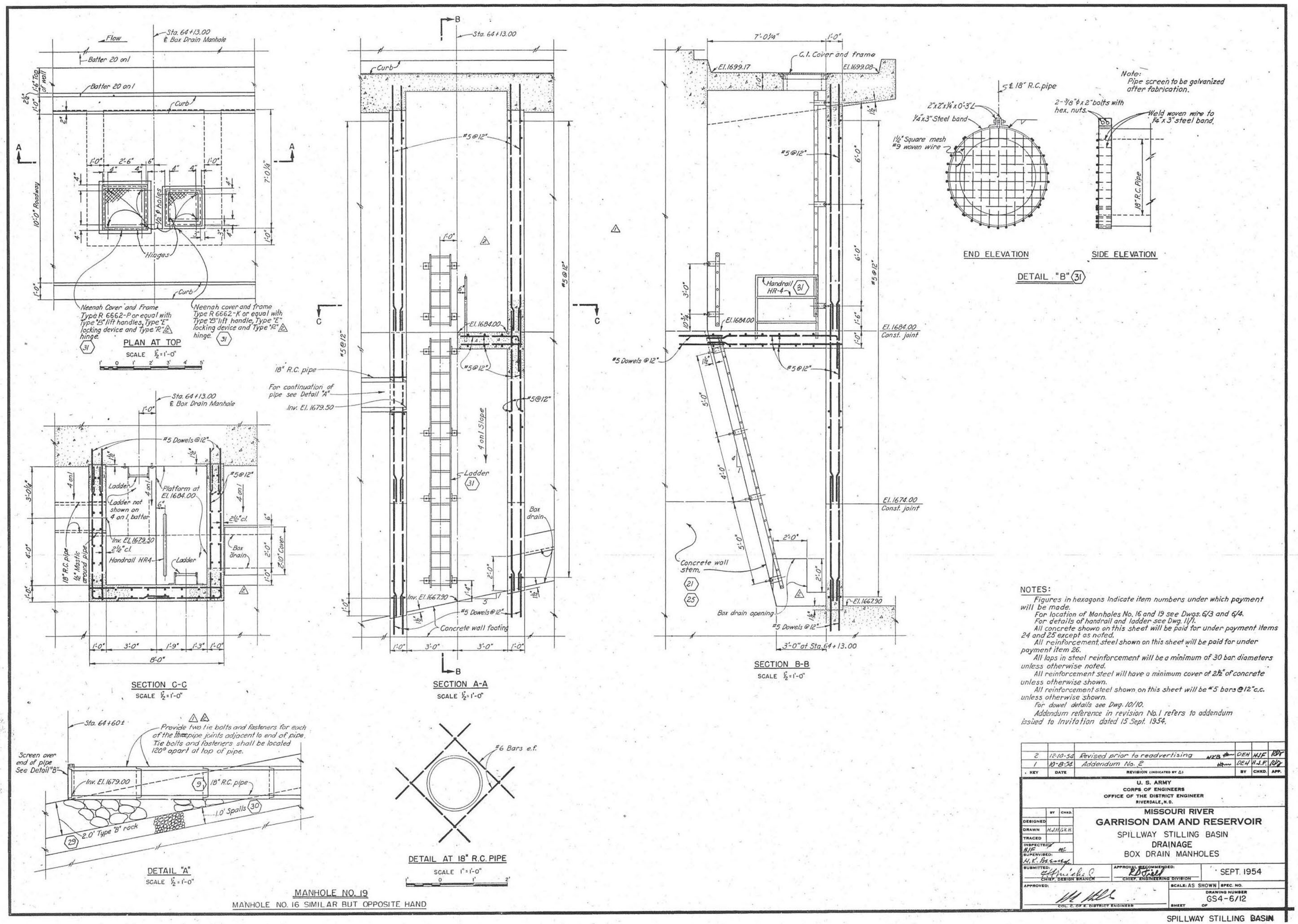


Figure 22. Box drain manholes original drawing (GS4, lower chute).

Figure 3-25: Auxiliary switchyard system existing conditions

<u>Original lines (GE10-7/42)</u>

- Vitrified clay, perforated pipe MH11 to MH14, and MH16 to 17

<u>Lines added prior to 1972</u>

- Perforated pipe to MH25
- Line to MH26 from MH16
  - Replaced 2013, GR57

<u>Lines first added 1972 (MGR86-88E)</u>

- MH29 with left and right laterals
  - Replaced 2003 with slotted steel, MGR283
- Line to MH27 and MH28 from MH16, and relief wells
  - Replaced 2010/2013, GR52/GR57

Figure 23. Powerhouse drainage network replacement history.

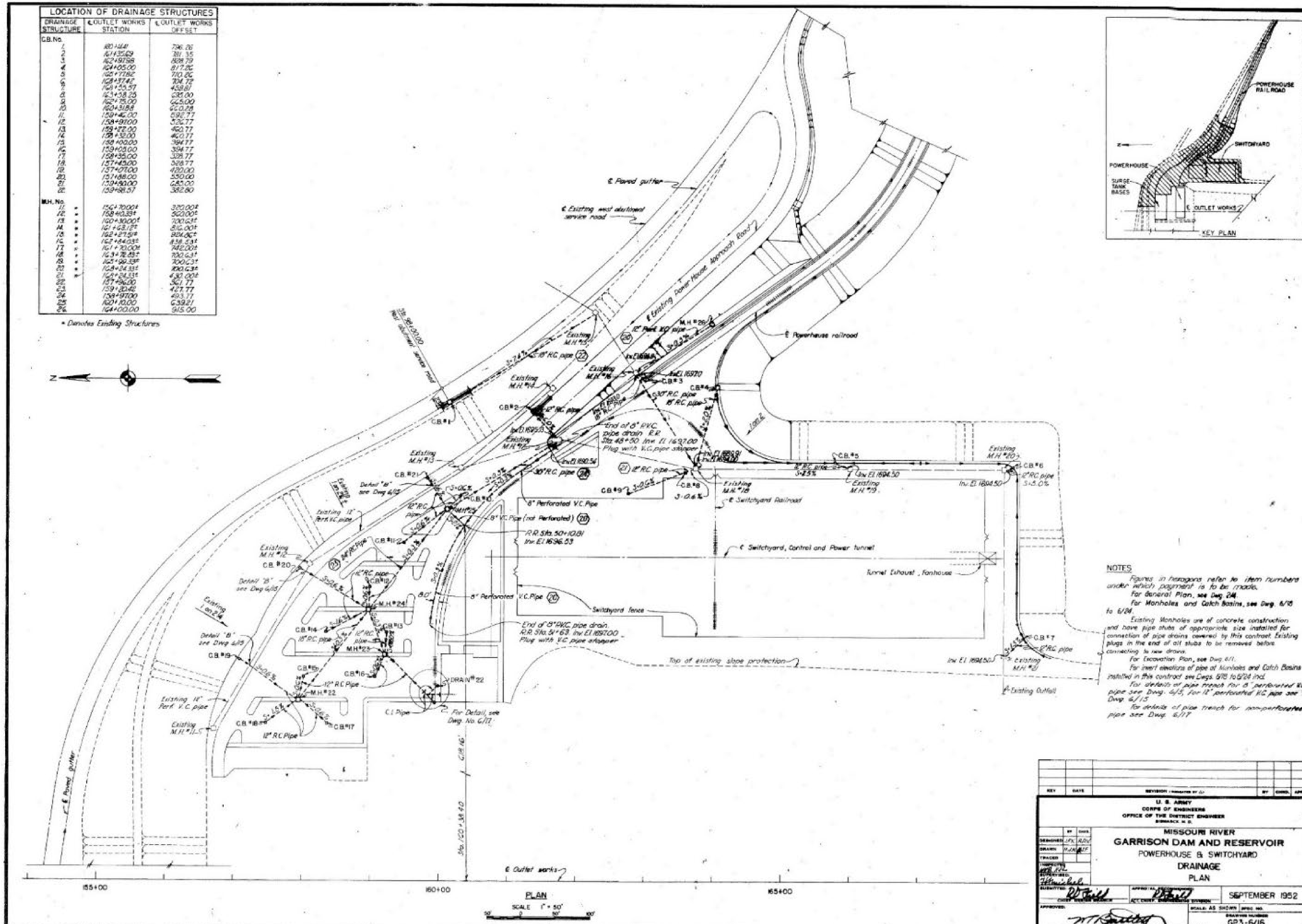


Figure 24. Powerplant Under Seepage Drain Network, original drawing.

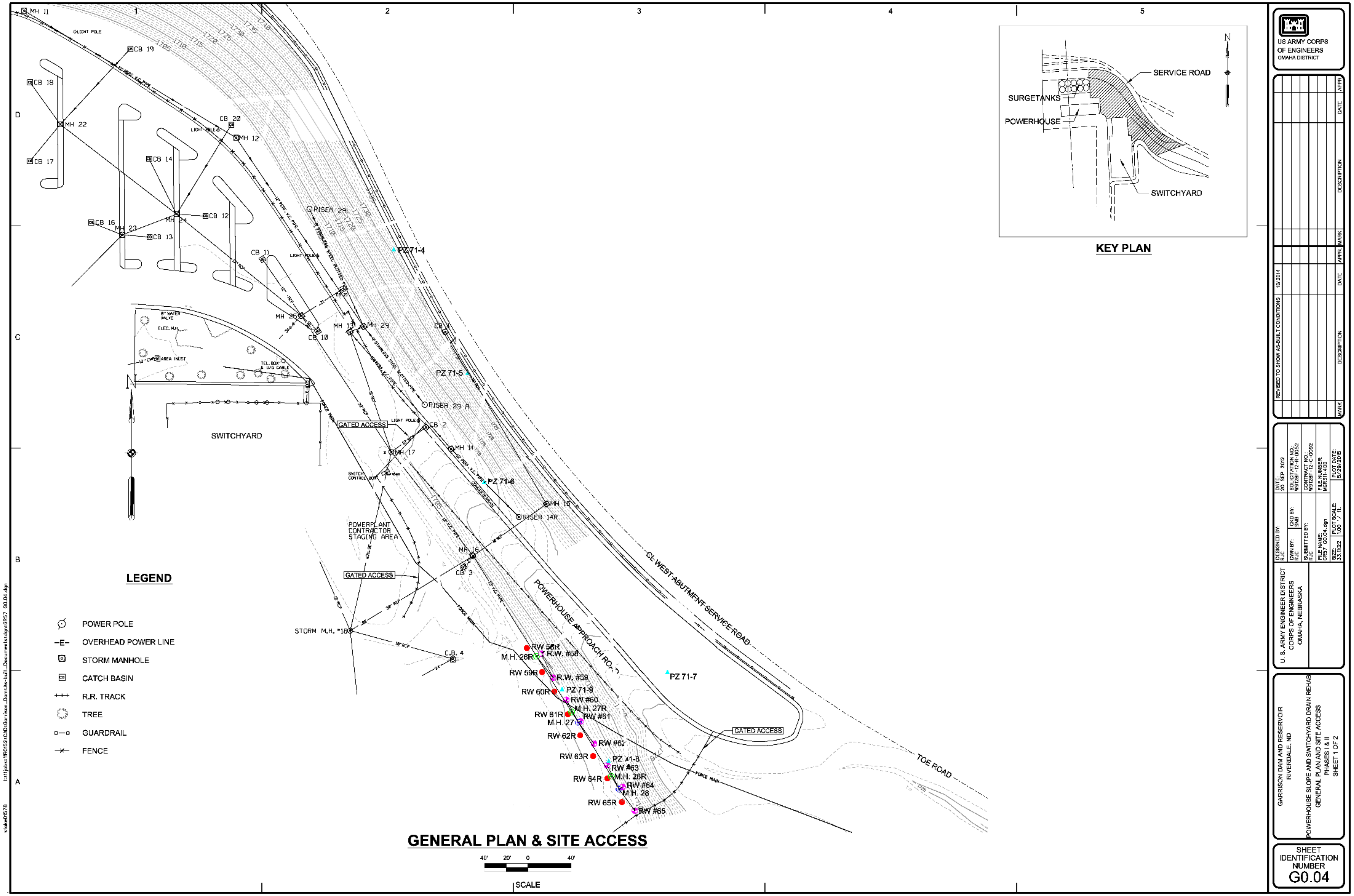
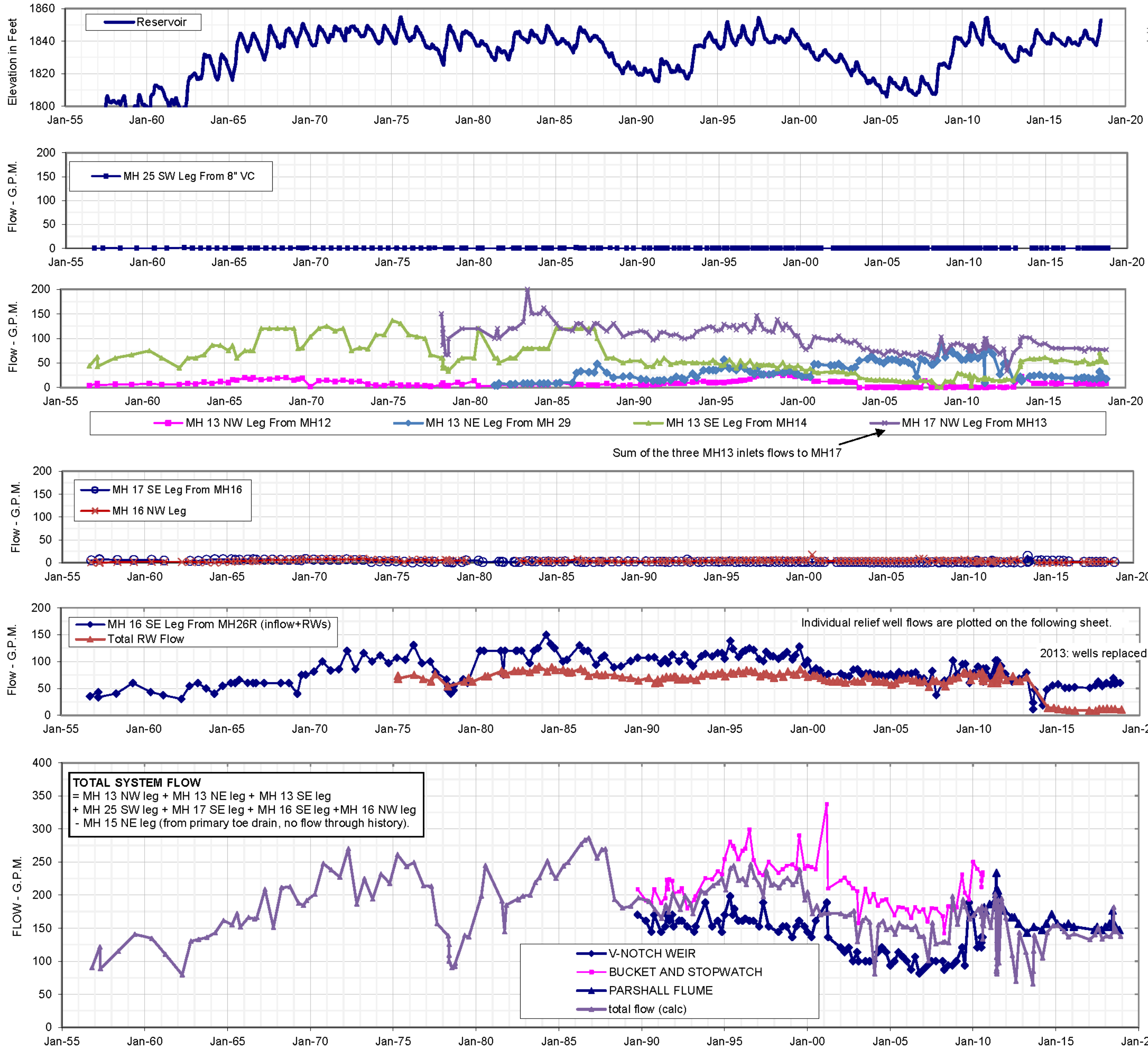
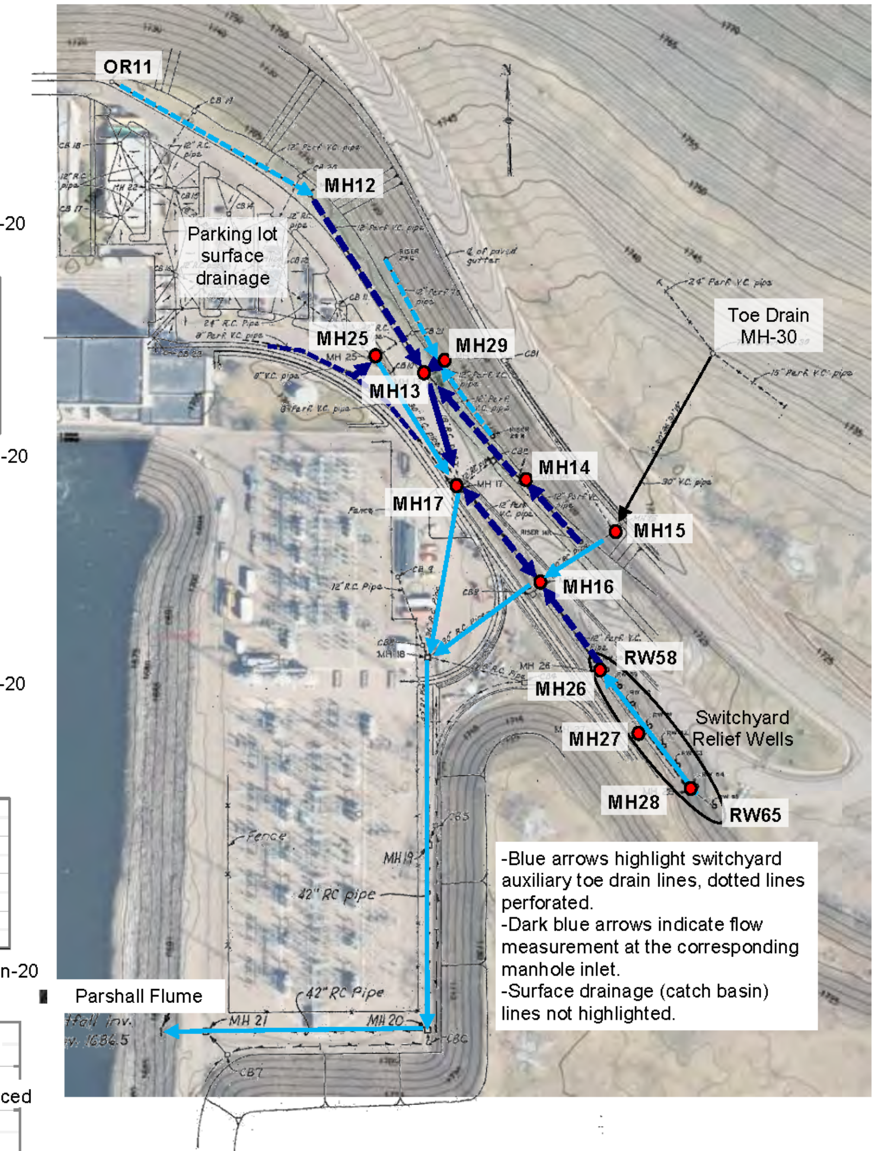


Figure 25. Powerhouse drainage map (2013)





Notes:

The Parshall flume measures the total flow collected in the Powerhouse & Switchyard Auxiliary Toe Drain System . The flume discharges into the left (east) side of the powerhouse tailrace. In 2010 the Parshall flume replaced a V-Notch Weir which was first installed in 1977.

A portion of the Primary Toe Drain system discharges into the Switchyard Auxiliary Toe Drain System from MH30. To date, zero flow has been recorded at this location.

The switchyard auxiliary toe drain system and relief wells primarily drain the west terrace gravels The majority of the system was cleaned or replaced in 2013.

MISSOURI RIVER
GARRISON DAM
POWERHOUSE & SWITCHYARD
AUILIARY TOE DRAIN
V-NOTCH WEIR
FLOW OBSERVATIONS

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Figure 26. Drainage Measurements - Powerhouse